Marketing Campaign Analytics

Project Report — Version 2

Updated with full EDA findings from new1_corrected.ipynb

Attribute	Value
Dataset	marketing_campaign_data.csv — 56,000 raw / 55,986 clean customer records
Countries	8 (Spain, Canada, Saudi Arabia, Australia, India, Germany, USA, Mexico)
Time Period	2012–2014 (customer enrollment dates)
Reference Date	2015-01-01 (used for Age, Tenure, Recency calculations)
Tools	Python, Pandas, NumPy, Seaborn, Plotly, MySQL, Streamlit, Apache Superset, Ollama
Deliverables	Python EDA notebook, SQL scripts, Streamlit app, Superset dashboard, Report

1. Executive Summary

A retail company operating across 8 countries ran 6 marketing campaigns (5 historical + 1 most recent) and collected detailed records of 56,000 customers covering demographics, spending behaviour, channel usage, and campaign acceptance. After cleaning, 55,986 valid customer records were analysed across 5 dimensions: campaign performance, customer segmentation, spending patterns, channel behaviour, and geography.

Top 6 Findings:

1. Income is the #1 predictor of both spend and campaign response — every 25K income band adds ~5–6% response probability.
2. The Premium segment (7.4% of customers) achieves 27.8% response rate — the highest ROI target.
3. Campaign 2 failed universally across all income and age bands at 1.4% — needs complete redesign.
4. 19,391 under-served customers visit 5+ times/month but never buy — the biggest untapped opportunity.
5. Catalog channels are used 52% more by high-value customers despite being the least-used overall.
6. The June 2014 enrollment spike is a data anomaly — Dt_Customer is not reliable for time-series analysis.

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2. Problem Statement

Management required a consolidated analytics solution to answer five core business questions:

1. Which customer segments have the highest response rate to campaigns — overall and per campaign?
2. How do spending patterns across products vary by age, income, marital status, and country?
3. Which channels are most used by high-value customers, and how often do they visit the website?
4. Are there specific segments that are under-served — high visits, low spending, low response?
5. What are the characteristics of ideal target customers for future campaigns?

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3. Data Understanding

3.1 Dataset Overview

Attribute	Value
Raw records	56,000 rows × 28 columns
Clean records	55,986 (14 income outliers removed = 0.025% data loss)
Engineered features	17 new columns added — Age, Total_Spend, Children, Age_Band, Income_Band, Primary_Segment etc.
Final dataset	55,986 rows × 45 columns
Missing values	None found in the raw dataset
Duplicate rows	None found

3.2 Data Quality Issues Found in EDA

Issue	Column	Severity	Resolution
Non-standard marital status values (YOLO, Absurd, Alone)	Marital_Status	Low	Mapped to 'Single' — confirmed these represent single-person households
Income extreme outlier (~258K)	Income	Medium	Removed 14 records beyond ±3σ — genuine anomalies not real customers
June 2014 enrollment spike — 3× normal volume	Dt_Customer	High	Confirmed data anomaly via Recency analysis — spike group has same recency as years-earlier customers, proving Dt_Customer was backdated. Field flagged as unreliable for time-series.
Recency is uniformly distributed 0–99	Recency	Medium	Unusual — real recency would be right-skewed. Consistent with Dt_Customer anomaly. Recency should not be used standalone for churn analysis.
Complain is 99:1 imbalanced	Complain	Low	Noted — too imbalanced for predictive modelling without resampling. Useful only as a filter.

4. Data Cleaning & Feature Engineering

4.1 Cleaning Steps

Step	Action	Result
Date conversion	Dt_Customer string → datetime; new DATE column Dt_Customer_Date added in MySQL	Enables temporal queries
Marital status	Mapped Absurd, YOLO, Alone → Single	Clean 5-category variable
Income outliers	Removed 14 records beyond ±3σ from mean (upper bound: ~157K)	55,986 records retained
Age outliers	No records outside 18–90 range found	No removal needed

4.2 Engineered Features

Feature	Formula	Purpose
Age	2015 − Year_Birth	Customer's age as of reference date
Total_Spend	Sum of all 6 Mnt* columns	Total wallet share (last 2 years)
Total_Purchases	Web + Catalog + Store + Deals	Total transaction count
Children	Kidhome + Teenhome	Total household dependants
Has_child	Children > 0 → Yes/No	Binary parental status for charts
Marital_Situation	Married/Together → 'In couple'; rest → 'Alone'	Simplified relationship status for sunburst charts
Total_Campaign_Accepted	Sum AcceptedCmp1–5 + Response (range 0–6)	Multi-campaign engagement score
Any_Campaign_Accepted	Total > 0 → 1/0	Ever-responded binary flag
Customer_Tenure_Days	Reference − Dt_Customer	Days as a customer
Customer_Tenure_Months	Tenure_Days // 30	Months as a customer
Age_Band	pd.cut(Age, 6 bins: 18–29 through 70+)	Age group for filtering and charts
Income_Band	pd.cut(Income, 5 bins: <25K through >100K)	Income tier for targeting

5. Exploratory Data Analysis — Key Findings

5.1 Univariate — Demographics

Variable	Key Finding	Business Implication
Age	Near-normal, centred 40–50 years. Young (<30) = 0.6% of base.	Working-age adults dominate. Young segment is too small for dedicated campaigns.
Income	Bell-shaped (after outlier removal), centred 50K–70K. Right tail = premium segment.	Meaningful premium income segment exists — worth separate targeting.
Total_Spend	Strongly right-skewed. Median 441, mean 640. 90th pct = 1,564.	Bi-modal population: budget shoppers + premium shoppers. Need two distinct strategies.
Education	Graduation 40.6%, PhD 22.5%, Master 18.8%, Basic 7.5%	Highly educated base — messaging can be sophisticated and data-driven.
Country	Spain 29.8%, Canada 19.2%, Saudi Arabia 15%, rest <11%	Two markets (Spain + Canada) = 49% of base. Country-specific campaigns warranted.
Marital Status	Together + Married ~56%, Single ~44%	Couple-households are the majority but single-person households spend more (see bivariate).
Children	68.2% have at least one child	Family-oriented offers have a broad addressable market.

5.2 Univariate — Spending

Category	Observation
Wines	Multi-modal distribution — two distinct buyer populations (light vs heavy). Largest revenue contributor (~50% of total).
Meat	Second-largest, right-skewed. Consistent buyers across demographics.
Fruits & Sweets	Narrow, unimodal — routine low-value purchases. No premium segment.
Fish & Gold	Log-normal — one main buyer group with an enthusiast tail.
Log transformation	Necessary to visualise buyer distributions — raw data is dominated by the large zero-spend group.

5.3 Univariate — Campaigns

Campaign	Rate	Total Accepted	Assessment
Response (Last Campaign)	14.75%	8,260	Best performing — strategy is improving
Campaign 1	13.44%	7,527	Good
Campaign 3	6.24%	3,495	Average
Campaign 4	5.68%	3,182	Below average
Campaign 5	4.57%	2,559	Below average
Campaign 2	1.44%	806	Universal failure — near-zero across all segments

5.4 Univariate — Channels

Channel	Avg Purchases	Distribution Shape	Key Insight
Store	4.71	Right-skewed, centred 3–6	Dominant channel for most customers
Web	4.25	Right-skewed, centred 3–6	Complementary to store, not substitute
Catalog	2.11	Heavily right-skewed	Low average but high tail = premium customer signal
Deals	2.17	Low, right-skewed	High deal usage signals price sensitivity
Web Visits	5.17	Left-skewed	Two distinct web engagement groups

Recency anomaly: Recency shows a near-uniform distribution 0–99 days — statistically unusual. Real customer databases show right-skewed recency (many recent buyers, fewer lapsed). This is consistent with the Dt_Customer data anomaly and suggests Recency may have been artificially capped or engineered. Do not use Recency as a standalone churn signal.

5.5 Bivariate — Correlation Analysis

Key correlations from the heatmap (Pearson r):

Relationship	Strength	Business Interpretation
Income ↔ MntWines	**+0.79 (Very High)**	Strong positive association — higher-income customers **tend to spend significantly more on wine**. Wine behaves like a premium category, but not exclusively (lower-income groups still participate).
Income ↔ MntMeatProducts	**+0.76 (Very High)**	Meat spending is also **strongly income-linked**, suggesting it is a higher-value category, though still broadly consumed across segments.
Income ↔ Total_Spend	**+0.57 (Moderate–High)**	Income is an **important driver of overall spending**, but not the only one — other factors (household size, preferences, promotions) also play a meaningful role.
Children ↔ Total_Spend	**−0.31 (Moderate Negative)**	Households with children **tend to spend less per customer**, possibly due to budget constraints or different spending priorities — but the effect is moderate, not dominant.
Children ↔ MntWines	**−0.23 (Weak–Moderate Negative)**	Customers with children **slightly reduce wine spending**, but the relationship is not strong — avoid over-segmentation purely on this factor.
Total_Campaign_Accepted ↔ Income	**+0.32 (Moderate)**	Higher-income customers are **somewhat more likely to accept campaigns**, but the relationship is not strong enough to rely on income alone for targeting.
AcceptedCmp* (all pairs)	**+0.05–0.20 (Weak)**	Campaign acceptance shows **weak positive consistency** — some tendency toward repeat acceptance, but not a strong loyalty signal. Likely influenced by campaign design and timing.
Recency ↔ anything	**≈ 0 (No Linear Relationship)**	Recency shows **no meaningful linear relationship** with other variables. This may indicate either low predictive value *or* a data quality / feature engineering issue.

5.6 Bivariate — Demographics vs Spend

Demographic	Key Finding
Marital Status	Divorced (754) and Widow (745) customers have the highest average spend — single-person households with more discretionary income. Married (505) spend least — family budget effect.
Education	Graduation (747) surprisingly spends more than PhD (576) — Graduation is larger and includes high-earning professionals without advanced degrees.
Income + Education	Clear gradient: Basic < 2n Cycle < PhD < Master < Graduation for income, but the difference is modest (~15% from Graduation to PhD).
Has Child	Child-free customers are over-represented in the >75K income bands. Parent customers cluster more in 25K–75K bands.

5.7 Bivariate — Product Spend by Demographics

Breakdown	Finding
By Income Band	Wines show the steepest income gradient — dominant in >100K. Meat follows. Fruits and Sweets are flat across income — they are routine purchases, not premium.
By Age Band	Spend peaks in the 40–49 age band. Wine increases consistently with age. 70+ shows a notable drop across all categories — likely retirement income constraints. Young (18–29) spend least across all categories.
Total contribution	Meat ~42% of all revenue. Meat ~39%. Fruits + Sweets + Fish + Gold ~ 19% combined. Wine and Meat = 80% of revenue — protecting these two categories is the #1 supply/margin priority.

5.8 Bivariate — Channels

Finding	Detail
Web and Store are complementary	More web purchases → more store purchases. No channel substitution effect. Omni-channel customers are of higher value.
Catalog = premium signal	Catalog purchases increase by 52% for high-value customers (Seg_High_Spender = 1). Largest uplift of any channel.
Store uplift for high-value	34% more store purchases for high-value customers vs average.
Deals decrease for high-value	−9% deal purchases for high-value customers. Premium customers are less price-sensitive.
Web visits conversion gap	Customers with 7–8 visits/month but only 1–2 purchases represent the under-served segment. High visit count + low purchase = conversion opportunity.

5.9 Bivariate — Campaign Acceptance

Dimension	Finding
By Income Band	Acceptance increases monotonically with income for ALL 6 campaigns. The >100K band accepts 2–4× more than the <25K band. Income-response relationship is robust across campaigns.
By Age Band	Acceptance generally increases with age. 50–69 is the sweet spot. 70+ shows a slight decline. Young (18–29) has the lowest acceptance across all campaigns.
Campaign 2 exception	Campaign 2 shows near-zero acceptance across ALL income bands AND all age bands — confirming it was a universal failure, not a targeting problem. The campaign itself was fundamentally flawed.
Segment uplift	The High Income segment shows ~80–100% more campaign acceptance than non-qualifying customers. High Spender follows. Family and Young show negative uplift — these segments are intrinsically less campaign-responsive.

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6. Rule-Based Customer Segmentation

Eight mutually exclusive segments were defined using business rules applied in priority order. The priority ordering ensures each customer receives their most commercially meaningful label when multiple rules apply simultaneously.

Priority	Segment	Rule	Count	Share	Avg Spend	Response Rate
1	Premium	High Spender AND High Income	4,135	7.4%	1,290+	27.8%
2	High Spender	Total_Spend > 1,564 (90th pct)	1,461	2.6%	1,800+	21.0%
3	High Income	Income > 75,000	15,633	27.9%	900	22.4%
4	Campaign Responder	Response = 1	3,302	5.9%	750	100%
5	Web Engaged	NumWebVisitsMonth > 5	19,722	35.2%	380	0%*
6	Family	Children > 0	8,864	15.8%	420	0%*
7	Young	Age < 30	322	0.6%	310	0%*

Priority	Segment	Rule	Count	Share	Avg Spend	Response Rate
8	Standard	All others	2,547	4.5%	290	0%*

** 0% reflects the primary segment bucket — responders are captured in Premium/High Income first due to priority ordering. The flag-level analysis (Seg_* columns) shows true response rates within each characteristic group.*

Segment Uplift on Campaign Acceptance: High Income segment members accept 80–100% more campaigns than non-qualifying customers. High Spender follows closely. Family and Young segments show NEGATIVE uplift — they require entirely different campaign types (value-driven, not premium) to achieve meaningful response.

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7. SQL Data Model

7.1 Schema Design

A single denormalised fact table (customers) holds all 45 columns. This design is appropriate for marketing analytics where queries are always customer-centric and no referential joins are needed. 7 indexes support fast dashboard filtering.

7.2 Analytical Views

View	Business Question	Key Technique
v_campaign_kpis	Which campaigns had the best/worst response rates?	UNION ALL of 6 SELECT statements; AVG on 0/1 flag columns
v_segment_kpis	How do the 8 segments compare across all KPIs?	GROUP BY Primary_Segment; 9 simultaneous AVG calculations
v_product_spend_by_segment	What product category does each segment buy most?	GROUP BY with 6 AVG calculations in one pass
v_channel_by_segment	Which purchase channel does each segment prefer?	GROUP BY with channel-specific AVGs + COUNT
v_country_summary	Which countries have the highest-value customers?	GROUP BY Country; ORDER BY avg_total_spend DESC
v_age_income_summary	How do age + income interact to affect spend/response?	Two-column GROUP BY (Age_Band × Income_Band cross-tab)

8. Dashboard Implementations

8.1 Streamlit Dashboard

The Streamlit app (app_mysql.py) provides 6 interactive tabs connected directly to MySQL via SQLAlchemy. A sidebar with 6 filters (Country, Education, Marital Status, Age Band, Income Band, Segment) applies dynamically to all charts simultaneously.

Tab	Key Visuals
Overview	5 KPI metric cards, segment pie chart, income band bar, education and marital summary tables
Campaign Analysis	Campaign response rate bar chart, segment response rate bar, income and age band line charts, multi-campaign acceptance table
Spending Patterns	Product category bar chart, box plots, age × product grouped bar, income vs spend scatter
Channel Analysis	Channel averages bar, high vs low spender comparison, web visits vs spend scatter, under-served segment profile
Customer Segments	Segment KPI table, radar chart (normalised KPIs), age and income box plots, ideal target profile
AI Data Explorer	Natural language prompt bar — Ollama/Groq generates pandas code → runs on filtered data → returns table or chart

8.2 Apache Superset Dashboard

The Superset dashboard connects to MySQL and provides an enterprise BI interface for non-technical stakeholders. It uses the 6 pre-built views as datasets, 13 charts, and 6 cross-chart filters.

9. Key Findings

Finding 1 — Income is the Single Strongest Predictor of Both Spend and Campaign Response

Response rate climbs monotonically with income across all 6 campaigns — from 4.6% in the <25K band to 27.8% in the >100K band. The income ↔ Total_Spend correlation is 0.79 — the

strongest bivariate relationship in the dataset. Every 25K increase in income band corresponds to approximately 5–6 percentage points of additional campaign response probability. Income outperforms age, education, marital status, and channel behaviour as a targeting variable.

Finding 2 — The Premium Segment Delivers the Highest ROI per Campaign Dollar

4,135 customers (7.4%) who qualify as both high-income and high-spenders achieve a 27.8% response rate and average spend exceeding 1,290. Catalog and store channels are significantly over-indexed for this group (+52% and +34% vs average respectively). They use deals 9% less than average — confirming they are quality-driven, not price-driven. Premium customers should receive premium channel investment, not discounts.

Finding 3 — Campaign 2 is a Universal Failure Across All Segments

Campaign 2's 1.4% acceptance rate is not a targeting problem — it is near-zero across every income band and every age band tested. This rules out demographic mismatch and points to a fundamental failure in the offer itself, the channel used, or the timing. The campaign must be retired entirely and rebuilt from scratch, not rerun with adjusted targeting.

Finding 4 — A Large Under-Served Segment Exists with High Web Engagement

19,391 customers visit the website 5+ times per month but have never responded to any campaign and spend below the median. Average age 41, average income 33,338. These customers are actively interested (evidenced by repeated web visits) but are not converting — they are price-sensitive and the current campaign offers are not designed for their income level. A specific entry-level offer at the 25K–50K income tier could unlock this group.

Finding 5 — Catalog is the Under-Invested Premium Channel

Catalog purchases increase by 52% for high-value customers compared to the overall average — the largest channel uplift of any channel. Despite being the least-used channel overall (avg 2.11 purchases), it is the preferred channel of the most valuable customers. Deal purchases decline by 9% for high-value customers. The current channel investment mix is misaligned with where premium customers actually spend — catalog deserves budget reallocation from deals.

Finding 6 — Multi-Campaign Loyalty Compounds with Income

Customers who accepted all 5 historical campaigns have average income 93,844 versus 48,913 for zero-acceptance customers — a 45K gap. Each additional accepted campaign corresponds to ~9,000 higher income. Campaign acceptors also show significantly higher Total_Spend. This compound effect suggests that maintaining engagement with premium customers across multiple campaigns builds long-term value, not just one-off response.

Finding 7 — Dt_Customer June 2014 Spike is a Data Anomaly

A 3× spike in recorded customer enrollments in June 2014 was investigated using Recency as a diagnostic. If these customers genuinely joined in June 2014, their Recency (days since last purchase as of Jan 2015) should be low. Instead, the spike group has identical Recency distribution to customers enrolled years earlier — confirming that Dt_Customer was backdated or incorrectly recorded for a large batch of records. The field should not be used for time-series cohort analysis without data remediation.

Finding 8 — Web and Store Channels are Complementary, Not Competing

Customers with higher NumWebPurchases also have higher NumStorePurchases — there is a positive correlation between the two channels. This disproves any assumption of digital-physical channel cannibalisation in this dataset. Omni-channel customers (active on both) are systematically higher-value. Encouraging web engagement should not come at the expense of in-store investment.

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10. Actionable Recommendations

Rec 1 — Target Premium + High Income Segments with Catalog and In-Store Campaigns

These 19,768 customers (35% of base) have response rates of 22–28%. Allocate 50%+ of campaign budget here. Use catalog and in-store channels — not digital push or discount offers. These customers respond to quality, exclusivity, and curated product recommendations, not price incentives.

Rec 2 — Retire Campaign 2 Completely

The 1.4% response rate is not an income or age targeting problem — it failed universally across all demographics. Do not attempt to rerun it with adjusted targeting. Audit the offer type, channel, and timing from scratch. A/B test a completely redesigned replacement against a holdout group before any full rollout.

Rec 3 — Launch an Entry-Level Offer for the Under-Served Web Segment

19,391 customers visit regularly but do not buy. A 10–15% discount or product bundle designed specifically for the 25K–50K income tier could convert even 10% of this group — generating ~1,900 new active buyers at very low acquisition cost. This should be a web-specific campaign given their digital engagement pattern.

Rec 4 — Reallocate Deal Budget to Catalog for Premium Segments

High-value customers use deals 9% less than average and catalog 52% more. Shifting budget from deals-and-discounts campaigns to catalog campaigns for the top 25% of customers by spend will improve both ROI and margin (fewer revenue-diluting discounts given to customers who would have bought anyway).

Rec 5 — Build a Formal Loyalty Escalation Programme

The compounding income-acceptance correlation is the most actionable long-term finding. Create a loyalty programme that rewards customers after each campaign acceptance — early access, curated recommendations, premium tier status. This reinforces the observed compound effect and is likely to increase Customer Lifetime Value for the Premium and High Income segments.

Rec 6 — Localise Campaigns for Spain and Canada

Spain (29.8%) and Canada (19.2%) together represent 49% of the customer base. Country-specific campaigns — wine-forward and premium-positioned for Spain, value-oriented and food-focused for Canada — will outperform a one-size-fits-all global campaign. These two markets are large enough to justify separate creative and channel strategies.

Rec 7 — Investigate and Remediate Dt_Customer Data Quality

The June 2014 spike and the uniform Recency distribution indicate structural data quality issues with enrollment date recording. Before any cohort-based analysis (e.g. first-year spend, early churn) can be done reliably, the source system needs to be audited and Dt_Customer values corrected where possible. Tag affected records with a data_quality_flag column for transparency.